

Eventual arithmetic sufficiency on every positive count ray

Extension of the positive-prefix completion argument

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Fix a positive integer vector $c = (c_1, \dots, c_n)$. Set

$$R(c) = \prod_{\substack{p \leq n \text{ prime} \\ |\{i: p|c_i\}| \geq p}} p.$$

The empty product is one. The subset-divisibility characterization says that the vector tc satisfies every necessary factorial divisibility condition if and only if $R(c) \mid t$. Indeed, at least p counts not divisible by p supply a violating p -subset. Conversely, if at most $p-1$ counts are p -units, every k -subset with $k \geq p$ has product valuation at least $k-p+1 \geq (k-1)/(p-1) \geq v_p(k!)$; for $k < p$ the requirement is empty. Applying this criterion after multiplication by t proves the claim.

Theorem 1. *For every positive integer vector c there is a finite threshold $N(c)$ such that, for every integer $t \geq N(c)$, a permutation-fair family with face-count vector tc exists if and only if $R(c) \mid t$. More specifically, some positive integer A has both Ac and $(A + R(c))c$ feasible. Every multiple of $R(c)$ at least $A(A - R(c))/R(c)$ is consequently a feasible dilation.*

The conclusion concerns sufficiently large dilations of a fixed profile. It does not bound the minimum total number of faces sharply, nor imply that the undilated arithmetic conditions are sufficient.

Use the reduced squarefree algebra and signed-signature theorem from `signed_fair_signatures.tex`. Put

$$H_c = \sum_i c_i X_i, \quad T_t^{(c)} = \exp(tH_c), \quad D = \text{lcm}(c_1, \dots, c_n).$$

The signature $T_t^{(c)}$ is exactly fairness with counts tc whenever it is represented by a positive word.

Lemma 2 (Weighted prefix completion). *Every positive word is a literal prefix of a fair positive word with count vector Ac for some positive integer A . More quantitatively, if $B \geq 1$ and the starting word has at most Bc_i occurrences of letter i , one may take*

$$A = B \prod_{k=2}^n (D^k + (n! - 1)D).$$

Proof. Append letters until the count vector is exactly Bc . Define weighted relabelings by

$$\psi_\sigma(X_i) = \frac{c_{\sigma(i)}}{c_i} X_{\sigma(i)} \quad (\sigma \in S_n).$$

These are a group action by algebra automorphisms, they fix H_c , and they are conjugate to ordinary relabelings by the diagonal change $X_i \mapsto c_i X_i$. Thus a homogeneous Lie polynomial of degree $k > 1$ invariant under every ψ_σ is zero, by the invariant-Lie lemma in the positive-completion note.

Suppose the current positive word w has

$$\log \text{sig}(w) = BH_c + E_k + \text{terms of degree greater than } k,$$

with all errors in degrees $2, \dots, k-1$ zero. Let δ_D be the dilation $X_i \mapsto DX_i$. For every nonidentity permutation σ , the image of w under $\delta_D \psi_\sigma$ is an ordinary positive word: replace each letter i by $Dc_{\sigma(i)}/c_i$ copies of $\sigma(i)$. All these multiplicities are positive integers.

Concatenate first D^k copies of w , and then these transformed words for every $\sigma \neq \text{id}$. The first block has logarithm

$$D^k BH_c + D^k E_k + \dots,$$

whereas the block for $\sigma \neq \text{id}$ has logarithm

$$DBH_c + D^k \psi_\sigma(E_k) + \dots.$$

All degree-one terms are proportional to H_c and commute. Any BCH bracket involving an error has degree at least $k+1$. The new degree- k error is therefore $D^k \sum_{\sigma \in S_n} \psi_\sigma(E_k) = 0$. The new first-degree multiplier is $B(D^k + (n! - 1)D)$, and the old word remains a literal prefix. Iterating for $k = 2, \dots, n$ gives the claim. $\square$

Lemma 3 (Clearing signed words on a prescribed ray). *For every signed group element g there is a positive integer A such that both $T_A^{(c)}$ and $gT_A^{(c)}$ are represented by positive words.*

Proof. Weighted completion supplies, for any positive signatures u, v , positive complements u', v' with $uu' = T_a^{(c)}$ and $vv' = T_b^{(c)}$. These fair signatures commute, so

$$u(u'T_b^{(c)}) = v(v'T_a^{(c)}) = T_{a+b}^{(c)}.$$

Hence positive signatures have common right multiples. As in the equal-count case, every signed element is a right fraction xy^{-1} of positive signatures. Complete $yz = T_A^{(c)}$; then $gT_A^{(c)} = xz$ is positive. $\square$

Proof of Theorem 1. For a prime p , multiplication of all counts by a dilation t either makes every count divisible by p (if $p \mid t$), or leaves unchanged the set of p -unit counts. The subset-divisibility characterization therefore gives exactly $R(c) \mid t$. In particular the signed-signature theorem gives $T_{R(c)}^{(c)}$ as a signed group element. Apply Lemma 3 to this element. The resulting positive signatures are $T_A^{(c)}$ and $T_{A+R(c)}^{(c)}$. Necessity gives $R(c) \mid A$.

Concatenations of these two fair words remain on the same fair ray. Writing $a = A/R(c)$, they realize every nonnegative integer combination of a and $a + 1$, multiplied by $R(c)$. Every integer at least $a(a - 1)$ is such a combination, proving the stated threshold. $\square$

Review status. The full weighted-power proof has an independent line-by-line mathematical review, and exact reduced-algebra verification has passed on six small count rays; see `../positive_saturation/verify_weighted_completion.py`. No machine-checked formal verification is claimed.